

My Personal Data Analytics Project

Amazon Sales Data Analytics Project

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Project Summary and Abstract

Item	Details
Dataset	Amazon Sale Report
Records	128,975
Final Records	128,746
Analysis Period	March–June 2022
Tools Used	Python, Pandas, NumPy, MySQL, Power BI, Excel
Project Type	End-to-End Data Analytics Portfolio Project

This project demonstrates a complete end-to-end data analytics workflow using Python, MySQL, and Power BI. Beginning with raw transactional data, the project covers data profiling, cleaning, validation, SQL-based business analysis, and interactive dashboard development to generate actionable business insights.

1. Project Objective

The objective of this project is to analyze Amazon sales data and derive meaningful business insights using Python, SQL, and Power BI. The project follows a complete data analytics workflow beginning with raw data collection, followed by data profiling, cleaning, validation, SQL-based business analysis, and finally interactive dashboard development.

The objective is to simulate an end-to-end Data Analyst workflow by transforming raw transactional data into a validated analytical dataset, extracting business insights using SQL, and presenting the findings through interactive Power BI dashboards.

2. Dataset Description

Dataset Name: Amazon Sale Report
Source: Kaggle - E-Commerce Sales Dataset
Dataset Size:

- Records: 128,975
- Columns: 24

The dataset primarily represents apparel sales transactions across multiple Indian states during the period from March to June 2022. The dataset contains Amazon sales transaction records including:

- Order Details
- Product Information
- Sales Amount

- Quantity Sold
- Product Category
- Product Size
- Shipment Information
- Fulfilment Method
- Courier Status
- Customer Location
- Promotional Campaign Information
- Business-to-Business Indicator

Key attributes include:

- Order ID
- Date
- Status
- SKU
- Category
- Size
- Quantity
- Amount
- City
- State
- Courier Status
- Fulfilment
- B2B

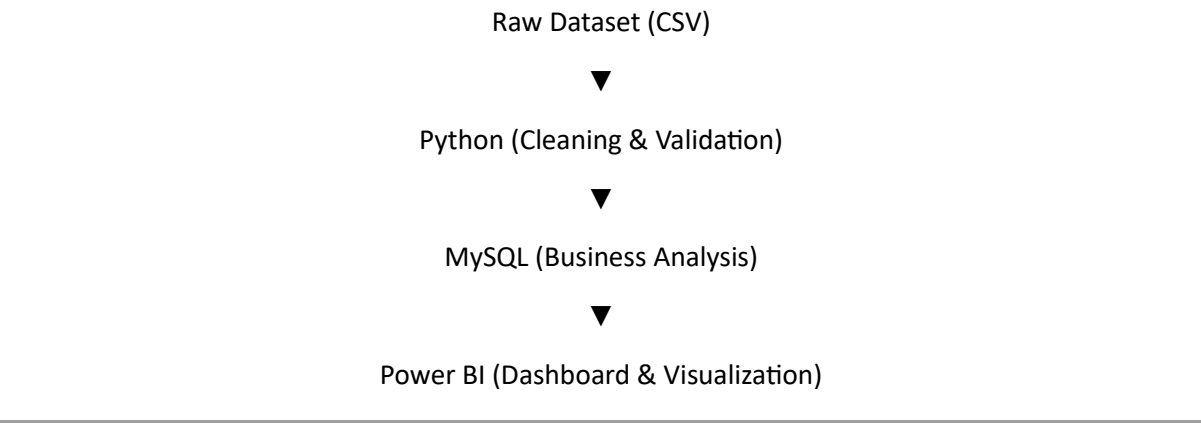
Dataset Selection Rationale

The Amazon Sale Report dataset was selected because it represents a realistic e-commerce business scenario containing transactional, product, fulfilment, and customer location information. The dataset provides sufficient complexity for performing end-to-end data analytics, including data cleaning, SQL-based business analysis, and dashboard development using Power BI.

3. Tools & Technologies Used

Tool	Purpose
Python	Data cleaning and preprocessing
Pandas	Data manipulation and validation
NumPy	Numerical operations
Jupyter Notebook	Exploratory analysis
MySQL	SQL analysis and KPI generation
Power BI	Interactive dashboard development and business visualization
Microsoft Excel	Initial data inspection

Project Workflow



4. Data Profiling

The dataset was initially explored using Python (Pandas) inside Jupyter Notebook.

The following profiling techniques were performed:

- Dataset structure inspection
- Data type validation
- Missing value analysis
- Duplicate detection
- Descriptive statistics
- Value distribution analysis
- Business rule validation

This stage helped identify several data quality issues that required cleaning before business analysis.

5. Data Quality Issues Identified

Missing Values

The dataset contained missing values across multiple columns. Most missing values occurred in a small number of key attributes and required investigation during the data cleaning process. The primary missing-value issues identified are summarized below.

Column	Missing Values
--------	----------------

fulfilled-by	89,698
--------------	--------

promotion-ids	49,153
---------------	--------

Unnamed: 22	49,050
-------------	--------

currency	7,795
----------	-------

Amount	7,795
--------	-------

Courier Status	6,872
----------------	-------

Data Type Issues

Several columns required formatting before analysis.

- Date stored as text.
- Postal Code imported as floating-point values.
- Unnamed column contained mixed data types.
- Boolean values required standardization before SQL import.

Duplicate Investigation

Potential duplicate records were investigated using **Order ID** and **SKU** combinations.

Although six duplicate Order ID–SKU combinations were identified, further inspection confirmed that there were **no completely duplicated rows** after the cleaning process.

6. Data Cleaning Process

Step 1: Removed Unnecessary Columns

The following columns were removed because they either contained redundant information or provided no analytical value:

- Unnamed: 22
- Sales Channel
- fulfilled-by
- currency

- promotion-ids
- index

Removing these columns reduced dataset complexity while preserving all information required for business analysis.

Step 2: Standardized Column Names

Column names were standardized to improve readability and SQL compatibility.

Examples:

- Order ID → order_id
- Courier Status → courier_status
- ship-service-level → ship_service_level
- Order_Date → order_date

A consistent snake_case naming convention was adopted throughout the project.

Step 3: Date Standardization

The Date column was converted from text format into a proper datetime format.

Original	Standardized
MM-DD-YY	YYYY-MM-DD

A new column named order_date was created for SQL analysis and Power BI reporting.

Analysis period:

- First Order: 31 March 2022
- Last Order: 29 June 2022

Step 4: Missing Amount Investigation

A total of **7,795** records contained missing Amount values.

Further investigation revealed:

- **7,566** records belonged to cancelled orders with Quantity = 0.
- **229** records belonged to non-cancelled orders despite having missing revenue.

Action Taken

- Cancelled orders were retained because they are operationally important.
- The 229 non-cancelled records were removed due to incomplete financial information.

Dataset size after cleaning:

Records: 128,746

Step 5: Quantity Validation

Quantity values were validated.

Findings:

- Minimum Quantity = 0
- Maximum Quantity = 8
- No negative quantities

All zero-quantity records corresponded to cancelled orders, confirming consistency with business rules.

Step 6: Revenue Validation

Amount values were analyzed.

Findings:

- Minimum Amount = 0
- Maximum Amount = 5,584

Approximately **1.82%** of the cleaned dataset contained an Amount value of ₹0 despite having shipment-related statuses. Since the underlying business reason could not be verified, these records were retained to preserve data integrity and avoid introducing bias into subsequent analyses.

Step 7: Fulfilment Validation

A relationship analysis between **Fulfilment** and **fulfilled-by** revealed that:

- Every Merchant fulfilment corresponded to Easy Ship.
- Every Amazon fulfilment contained null values in fulfilled-by.

Since fulfilled-by added no new business information, the column was removed.

Step 8: Country Validation

The **ship_country** column was analyzed.

Findings:

- Nearly every valid record contained **IN (India)**.

The small number of missing values was filled with **IN**, ensuring consistency while maintaining business meaning.

Step 9: Courier Status Validation

Missing Courier Status values were investigated.

It was observed that all missing Courier Status values corresponded to **Cancelled** orders.

These missing values were replaced with **Cancelled**, improving data consistency.

Step 10 – Geographic Data Validation

During exploratory SQL analysis, additional data quality issues were identified in the **ship_state** column.

Several spelling variations and naming inconsistencies (such as *Orissa* and *Odisha*) were standardized where the correct value could be determined with confidence.

After standardization, **28 records contained missing (blank) state values**, while **5 records contained ambiguous or non-standard geographic entries** (such as abbreviated or location-specific values). Since the correct state could not be verified for these records, they were retained without modification to preserve data integrity and avoid introducing unsupported assumptions into analysis.

This iterative validation process ensured that the dataset used for SQL analysis and Power BI visualization was both accurate and reliable while maintaining authenticity of original source data.

```
df_clean.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 128746 entries, 0 to 128745
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   order_id              128746 non-null object
1   date                  128746 non-null object
2   status                128746 non-null object
3   fulfilment            128746 non-null object
4   ship_service_level    128746 non-null object
5   style                 128746 non-null object
6   sku                   128746 non-null object
7   category              128746 non-null object
8   size                  128746 non-null object
9   asin                  128746 non-null object
10  courier_status         128746 non-null object
11  qty                    128746 non-null int64
12  amount                 121180 non-null float64
13  ship_city              128713 non-null object
14  ship_state             128713 non-null object
15  ship_postal_code       128713 non-null float64
16  ship_country           128746 non-null object
17  b2b                    128746 non-null bool
18  order_date             128746 non-null datetime64[ns]
dtypes: bool(1), datetime64[ns](1), float64(2), int64(1), object(14)
memory usage: 17.8+ MB
```

Figure 1. Final cleaned analytical dataset after data preprocessing in Python.

7. Final Analytical Dataset

Final Dataset Size

- Records: 128,746
- Columns: 19

After removing six non-essential columns, standardizing the dataset, and creating the `order_date` field, the final analytical dataset contained **19 validated business-relevant attributes** prepared for SQL analysis and Power BI dashboard development.

No.	Column	Description
1	<code>order_id</code>	Unique order identifier
2	<code>date</code>	Original order date (text format)
3	<code>status</code>	Current order status

No.	Column	Description
4	fulfilment	Fulfilment method
5	ship_service_level	Shipping service level
6	style	Product style/design
7	sku	Stock Keeping Unit
8	category	Product category
9	size	Product size
10	asin	Amazon Standard Identification Number
11	courier_status	Courier delivery status
12	qty	Quantity ordered
13	amount	Sales amount (₹)
14	ship_city	Destination city
15	ship_state	Destination state
16	ship_postal_code	Postal code
17	ship_country	Destination country
18	b2b	Business-to-Business indicator
19	order_date	Standardized date used for analysis

This finalized analytical dataset served as the single source of truth for all subsequent SQL analysis and Power BI dashboard development.

8. SQL Analysis

After completing data cleaning and validation, the analytical dataset was imported into MySQL for business-oriented SQL analysis. The SQL phase focused on answering key business questions through structured queries, generating KPIs, and identifying patterns in product performance.

The SQL analysis was organized into multiple business-focused sections, each designed to answer a specific analytical question using structured SQL queries.

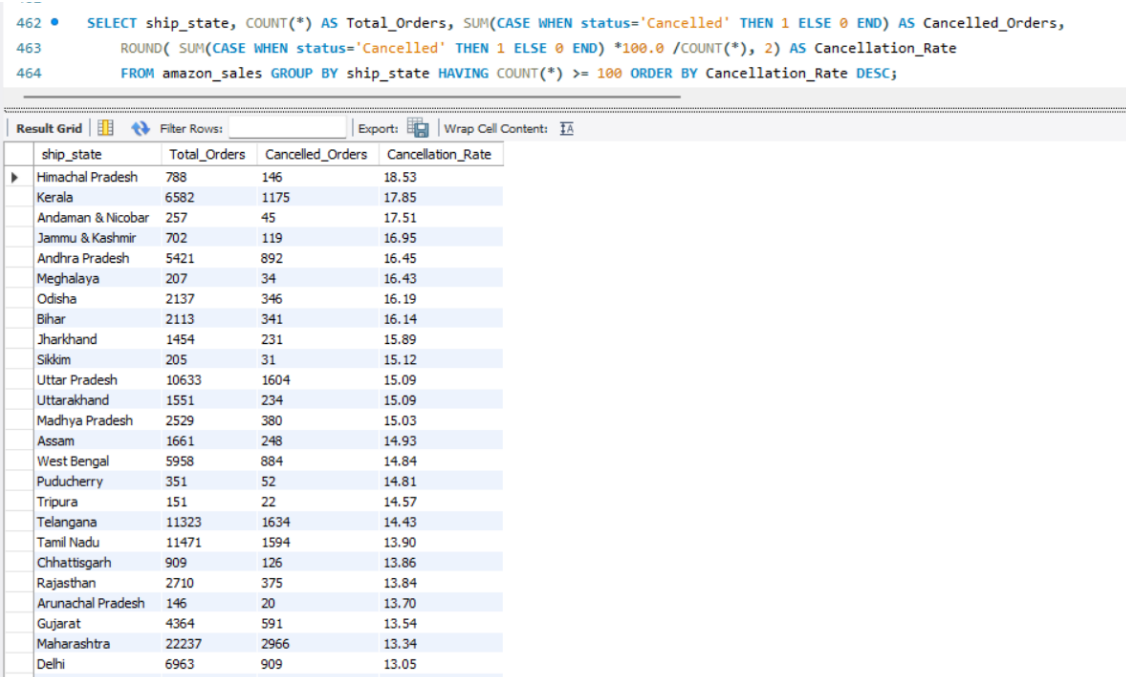


Figure 2. SQL analysis identifying cancellation rates across different states.

8.1. Data Validation

Before beginning analysis, the imported data was validated.

Validation results:

Metric	Result
Total Records	128,746
Unique Orders	120,255
Unique Products (SKU)	7,185
Analysis Period	31 March 2022 – 29 June 2022

This validation confirmed that the SQL dataset matched the cleaned dataset generated in Python.

8.2. Executive KPI Analysis

Executive KPIs were developed to provide a high-level overview of business performance and establish baseline metrics for subsequent analysis.

Key Findings

KPI	Result
Total Revenue	₹78,592,678.30
Total Orders	120,255

KPI	Result
Total Quantity Sold	116,483 Units
Average Order Value	₹648.56
Cancellation Rate	14.24%

Key Insights

- The business generated approximately **₹78.59 million** during the analysis period.
- More than **120 thousand** customer orders were processed.
- Approximately **one in every seven orders** was cancelled.
- The average revenue generated per successful order was **₹648.56**.

8.3. Product Performance Analysis

Product performance analysis was conducted to identify the company's highest-performing product categories, bestselling products, pricing patterns, and customer size preferences.

Revenue by Category

The Set category generated approximately **₹39.20 million**, accounting for nearly **50% of the total sales revenue**. This indicates that Set products are the primary revenue driver and should receive strategic attention in inventory planning and marketing.

Units Sold by Category

The **Set** category also recorded the highest unit sales, closely followed by **Kurta**, indicating that higher revenue was largely driven by strong sales volume.

Average Selling Price

The Set category combined both high sales volume and the highest average selling price among major categories, making it the strongest overall contributor to business revenue.

Top Revenue-Generating Products

Multiple variants of the same product families consistently appeared among the top revenue-generating SKUs, indicating strong and sustained customer demand for these product lines.

Size Analysis

Medium (M), Large (L), and Extra Large (XL) represented the highest-selling sizes, suggesting that inventory allocation should prioritize these sizes to reduce stock shortages and improve customer satisfaction.

Analysis	Key Finding
Highest Revenue Category	Set

Analysis	Key Finding
Highest Units Sold	Set
Highest Average Price	Set
Best-selling Size	M
Top Revenue SKU	J0230-SKD-M

8.4. Geographic Analysis

Geographic analysis was performed to identify regional sales trends, customer demand, and operational performance across different states and cities. This analysis helps businesses understand where revenue is generated, identify high-demand markets, and recognize regions requiring operational improvements.

Revenue by State

State-wise revenue analysis revealed that **Maharashtra** generated the highest revenue of approximately **₹13.34 million**, followed by **Karnataka** (₹10.48 million) and **Telangana** (₹6.92 million). These three states represent the company's strongest regional markets and contribute a significant portion of the overall sales revenue.

The close relationship between revenue and order volume indicates that these states are driven primarily by strong customer demand rather than unusually high product prices.

Orders by State

Maharashtra recorded the highest number of customer orders (**20,780**), followed by Karnataka (**16,182**) and Tamil Nadu (**10,519**). The order distribution closely mirrors the revenue distribution, suggesting that higher regional revenue is largely driven by greater sales volume rather than pricing differences.

These findings indicate that inventory planning, warehouse allocation, and marketing investments should prioritize the highest-order states to support sustained business growth.

Revenue by City

At the city level, **Bengaluru** generated the highest revenue (**₹7.26 million**), followed by **Hyderabad** (**₹5.60 million**), **Mumbai** (**₹4.29 million**), **New Delhi** (**₹3.95 million**), and **Chennai** (**₹3.61 million**).

The analysis shows that major metropolitan cities account for a substantial proportion of total revenue, making them strategic locations for promotional campaigns, inventory planning, and logistics optimization.

Orders by City

Bengaluru also recorded the highest number of customer orders (**11,096**), followed by Hyderabad (**8,369**), Mumbai (**6,572**), New Delhi (**5,948**), and Chennai (**5,726**).

The similarity between the revenue and order rankings further confirms that sales performance is primarily influenced by customer demand in densely populated metropolitan regions.

Cancellation Rate by State

State-level cancellation analysis revealed noticeable regional differences in operational performance.

Among states with significant order volumes, **Himachal Pradesh** recorded the highest cancellation rate (**18.53%**), followed by **Kerala (17.85%)** and **Andhra Pradesh (16.45%)**. These values exceed the project's overall cancellation rate of **14.24%**, indicating potential operational challenges such as inventory shortages, delivery issues, or customer-related factors.

In contrast, states such as **Karnataka (12.96%)**, **Punjab (12.90%)**, and **Haryana (12.77%)** maintained cancellation rates below the overall average, suggesting comparatively efficient order fulfilment.

These findings provide valuable guidance for identifying regions where operational improvements could reduce cancellations and improve customer satisfaction.

Customer Segment Analysis (B2B vs B2C)

The customer segmentation analysis showed that **Business-to-Consumer (B2C)** transactions accounted for the overwhelming majority of both orders and revenue.

Although **Business-to-Business (B2B)** orders represented only a small proportion of total transactions, they generated a higher average order value (**₹701.33**) compared with **₹648.19** for B2C orders. This suggests that while B2B contributes relatively little to the overall business volume, it represents a potentially valuable market segment for future expansion.

Key Insights

- Maharashtra emerged as the highest revenue-generating state, while Bengaluru was the highest-performing city.
- Revenue distribution closely follows order volume, indicating that regional sales are primarily driven by customer demand.
- Metropolitan cities contribute a significant share of the company's overall revenue and order volume.
- Certain states exhibit cancellation rates above the project average, highlighting opportunities for operational improvements.
- Although B2B sales account for a relatively small proportion of transactions, they achieve a higher average order value than B2C sales.

Business Recommendations

Based on the geographic analysis, the following recommendations are proposed:

- Prioritize inventory allocation and warehouse capacity in the highest revenue-generating states, particularly Maharashtra, Karnataka, and Telangana.
- Focus regional marketing campaigns on high-performing metropolitan cities such as Bengaluru, Hyderabad, Mumbai, and Chennai to maximize customer reach.
- Investigate operational issues contributing to high cancellation rates in states such as Himachal Pradesh, Kerala, and Andhra Pradesh to improve fulfilment performance.

- Explore opportunities to expand B2B partnerships through targeted sales initiatives, as B2B customers demonstrate higher average order values.
- Continue monitoring regional sales performance through interactive Power BI dashboards to support data-driven decision-making.

Section Summary

Analysis	Key Finding
Highest Revenue State	Maharashtra
Highest Revenue City	Bengaluru
Highest Orders State	Maharashtra
Highest Orders City	Bengaluru
Highest Cancellation Rate	Himachal Pradesh
Higher Average Order Value	B2B

8.5. Fulfilment & Logistics Analysis

Fulfilment and logistics analysis was conducted to evaluate the operational performance of different fulfilment methods and shipping services. The analysis focused on revenue contribution, order volume, average order value, cancellation rates, and courier performance to identify opportunities for improving operational efficiency and customer satisfaction.

Revenue by Fulfilment

The analysis revealed that **Amazon Fulfilment** generated the highest revenue of approximately **₹54.32 million**, accounting for nearly **69% of the total revenue**, while **Merchant Fulfilment** generated **₹24.27 million**. This indicates that Amazon's fulfilment network plays the dominant role in the company's sales operations and contributes significantly to overall business performance.

Orders by Fulfilment

Amazon Fulfilment processed **89,480 orders**, more than twice the number processed by Merchant Fulfilment (**39,266 orders**). The distribution of orders closely aligns with the revenue analysis, confirming that Amazon Fulfilment is responsible for handling the majority of customer transactions.

Average Order Value by Fulfilment

The average order values for both fulfilment methods were almost identical. Amazon Fulfilment recorded an average order value of **₹649.48**, while Merchant Fulfilment recorded **₹646.51**. This suggests that the difference in total revenue is primarily driven by order volume rather than differences in individual order values.

Cancellation Rate by Fulfilment

Merchant Fulfilment recorded a cancellation rate of **17.47%**, considerably higher than the **12.82%** observed for Amazon Fulfilment.

This indicates potential operational challenges within the Merchant fulfilment process, such as inventory management, order processing, or fulfilment efficiency.

Reducing cancellations within Merchant Fulfilment could significantly improve realized revenue and enhance customer satisfaction.

Revenue by Shipping Service

The **Expedited** shipping service generated approximately **₹54.29 million** in revenue, while the **Standard** shipping service generated **₹24.31 million**. Similar to the fulfilment analysis, Expedited shipping contributes the majority of business revenue due to its substantially higher order volume.

Orders by Shipping Service

Expedited shipping processed **88,520 orders**, compared with **40,226 orders** processed through Standard shipping. The results indicate that Expedited shipping is the company's primary delivery method and is preferred for the majority of customer orders.

Cancellation Rate by Shipping Service

Standard shipping recorded a cancellation rate of **17.18%**, compared with **12.90%** for Expedited shipping. The higher cancellation rate associated with Standard shipping suggests opportunities to improve logistics efficiency, inventory availability, and delivery reliability.

Courier Status Distribution

Courier status analysis showed that **85.04%** of all orders were successfully shipped, demonstrating strong operational performance. Approximately **5.33%** of orders had no assigned courier, **5.09%** remained unshipped, and **4.54%** were cancelled.

Although the majority of orders progressed successfully through the fulfilment process, monitoring orders without assigned couriers and reducing the number of unshipped orders could further improve operational efficiency.

Fulfilment and Shipping Service Comparison

A combined analysis of fulfilment method and shipping service revealed that **Merchant Fulfilment using Standard shipping** recorded the highest cancellation rate (**17.47%**). In contrast, **Amazon Fulfilment with Expedited shipping**, despite processing the largest number of orders, maintained a considerably lower cancellation rate (**12.90%**). Amazon Fulfilment with Standard shipping achieved the lowest cancellation rate (**5.00%**), although it represented only a small proportion of total orders.

These findings demonstrate that operational performance varies considerably across fulfilment and shipping combinations, highlighting specific areas where process improvements could reduce cancellations and improve service quality.

Key Insights

- Amazon Fulfilment generated nearly **69% of total revenue** and processed approximately **70% of all customer orders**.
- Average order values were nearly identical across fulfilment methods, indicating that revenue differences are primarily driven by order volume.
- Merchant Fulfilment recorded a noticeably higher cancellation rate than Amazon Fulfilment.

- Expedited shipping generated higher revenue, processed more orders, and maintained a lower cancellation rate than Standard shipping.
- Merchant Fulfilment combined with Standard shipping exhibited the highest cancellation rate, indicating an opportunity for operational improvement.
- More than **85% of all orders** were successfully shipped, reflecting generally efficient fulfilment operations.

Business Recommendations

Based on the fulfilment and logistics analysis, the following recommendations are proposed:

- Continue prioritizing Amazon Fulfilment for high-demand products to leverage its superior operational performance.
- Investigate the causes of higher cancellation rates within Merchant Fulfilment and implement corrective measures related to inventory management and order processing.
- Improve the efficiency of Standard shipping operations to reduce cancellations and enhance delivery reliability.
- Monitor orders without assigned couriers to minimize fulfilment delays and improve customer experience.
- Continuously track fulfilment and shipping performance through interactive Power BI dashboards to identify operational trends and support data-driven decision-making.

Section Summary

Analysis	Key Finding
Highest Revenue Fulfilment	Amazon Fulfilment (₹54.32M)
Highest Order Volume	Amazon Fulfilment (89,480 Orders)
Highest Average Order Value	Amazon Fulfilment (₹649.48)
Lowest Cancellation Rate	Amazon Fulfilment (12.82%)
Highest Revenue Shipping Service	Expedited Shipping (₹54.29M)
Highest Order Volume Shipping Service	Expedited Shipping (88,520 Orders)
Lowest Cancellation Rate Shipping Service	Expedited Shipping (12.90%)
Highest Cancellation Combination	Merchant Fulfilment + Standard Shipping (17.47%)

8.6. Time Series Analysis

Time series analysis was conducted to examine how sales performance changed over the analysis period. Monthly and daily trends were analyzed to understand revenue fluctuations, customer purchasing behaviour, cancellation patterns, and overall business performance over time.

Monthly Revenue Trend

Monthly revenue analysis showed that **April generated the highest revenue (₹28.84 million)**, followed by **May (₹26.23 million)** and **June (₹23.43 million)**. Although March recorded only ₹101.68 thousand, it contained data for only a single day (31 March) and was therefore not considered representative for monthly comparison.

The results indicate a gradual decline in revenue from April to June.

Monthly Order Trend

Customer order volume followed a pattern similar to revenue. **April recorded the highest number of orders (45,806)**, followed by **May (39,180)** and **June (35,111)**.

The consistent decline in both revenue and order volume suggests reduced customer purchasing activity during the later months of the analysis period.

Monthly Average Order Value

Despite the decline in revenue and order volume, the average order value remained relatively stable throughout the analysis period. April recorded an average order value of ₹626.00, while May and June recorded ₹663.36 and ₹661.48, respectively.

This indicates that the reduction in revenue was primarily driven by fewer customer orders rather than lower-value purchases.

Daily Revenue Trend

Daily revenue analysis revealed noticeable fluctuations throughout the analysis period, with several high-performing sales days during April and early May. Revenue remained relatively stable during most of the period but showed a gradual decline toward the end of June.

The highest daily revenue was recorded on **4 May 2022**, generating approximately ₹1.21 million in sales.

Daily Order Trend

Daily order volume closely followed the revenue trend. Customer orders remained relatively consistent during April and early May before gradually decreasing during June.

This consistency suggests a strong relationship between daily order volume and daily revenue generation.

Revenue by Day of the Week

Revenue analysis by weekday showed that **Sunday generated the highest revenue (₹12.03 million)**, followed by **Tuesday (₹11.49 million)** and **Saturday (₹11.34 million)**. **Thursday generated the lowest revenue (₹10.36 million)**.

The relatively balanced distribution indicates that customer purchases occurred consistently throughout the week, although weekends demonstrated slightly stronger sales performance.

Orders by Day of the Week

Sunday also recorded the highest customer order volume (**18,178 orders**), while Thursday processed the fewest orders (**15,990 orders**).

The similarity between weekday revenue and order trends further confirms that revenue fluctuations were primarily influenced by customer order volume.

Monthly Cancellation Trend

Monthly cancellation analysis showed relatively stable cancellation rates throughout the analysis period. April recorded the highest cancellation rate (**14.57%**), while May (**14.00%**) and June (**14.09%**) remained almost identical.

The consistency of cancellation rates suggests that operational performance remained stable throughout the observed period.

Highest Revenue Days

The highest revenue day during the analysis period was **4 May 2022**, generating approximately **₹1.21 million** in revenue. Several other high-performing days occurred during late April and early May, indicating a period of particularly strong sales activity.

Key Insights

- April generated the highest monthly revenue and order volume during the analysis period.
- Revenue and customer orders gradually declined from April through June.
- Average order value remained relatively stable despite declining order volume.
- Sunday generated the highest revenue and customer order volume.
- Thursday recorded the lowest sales performance among weekdays.
- Cancellation rates remained stable across all months, indicating consistent operational performance.
- Early May represented the strongest sales period within the dataset, with multiple days exceeding **₹1 million** in daily revenue.

Business Recommendations

Based on the time series analysis, the following recommendations are proposed:

- Investigate the factors contributing to declining revenue and customer orders after April to identify potential changes in customer demand or market conditions.
- Schedule promotional campaigns and marketing activities during weekdays with comparatively lower sales, particularly Thursdays, to improve weekly sales distribution.
- Capitalize on stronger weekend customer activity by increasing promotional offers, inventory availability, and marketing efforts during Saturdays and Sundays.
- Continue monitoring monthly cancellation rates to maintain operational consistency and identify any future deviations.
- Utilize daily sales trends for demand forecasting and inventory planning to improve supply chain efficiency.

Section Summary

Analysis

Highest Monthly Revenue

Key Finding

April (₹28.84M)

Highest Monthly Orders

April (45,806 Orders)

Highest Average Order Value

May (₹663.36)

Highest Revenue Weekday

Sunday (₹12.03M)

Highest Order Volume Weekday

Sunday (18,178 Orders)

Lowest Revenue Weekday

Thursday (₹10.36M)

Highest Daily Revenue

4 May 2022 (₹1.21M)

Highest Monthly Cancellation Rate

April (14.57%)

9. Power BI Dashboard Development

9.1 Executive Overview

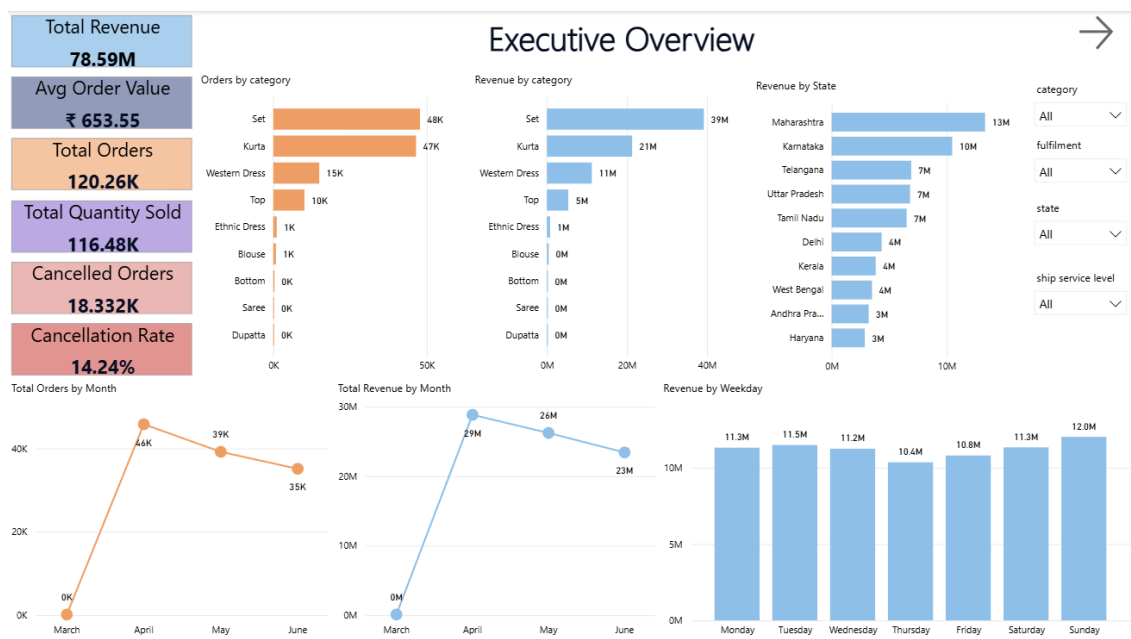


Figure 3. Executive Overview Dashboard developed in Power BI.

The Executive Overview dashboard provides a high-level summary of overall business performance using interactive visualizations and key performance indicators. It was designed to help stakeholders quickly understand sales performance and monitor business health.

The dashboard includes:

- KPI cards displaying Total Revenue, Total Orders, Total Quantity Sold, Average Order Value, and Cancellation Rate.
- Monthly revenue and order trends to identify sales patterns.

- Revenue and order distribution by product category.
- Revenue analysis by weekday.
- Top-performing states based on revenue.
- Interactive slicers for Category, State, Fulfilment Method, and Shipping Service Level to enable dynamic analysis.

This dashboard enables users to monitor overall business performance and quickly identify major revenue drivers.

9.2 Geographic Analysis

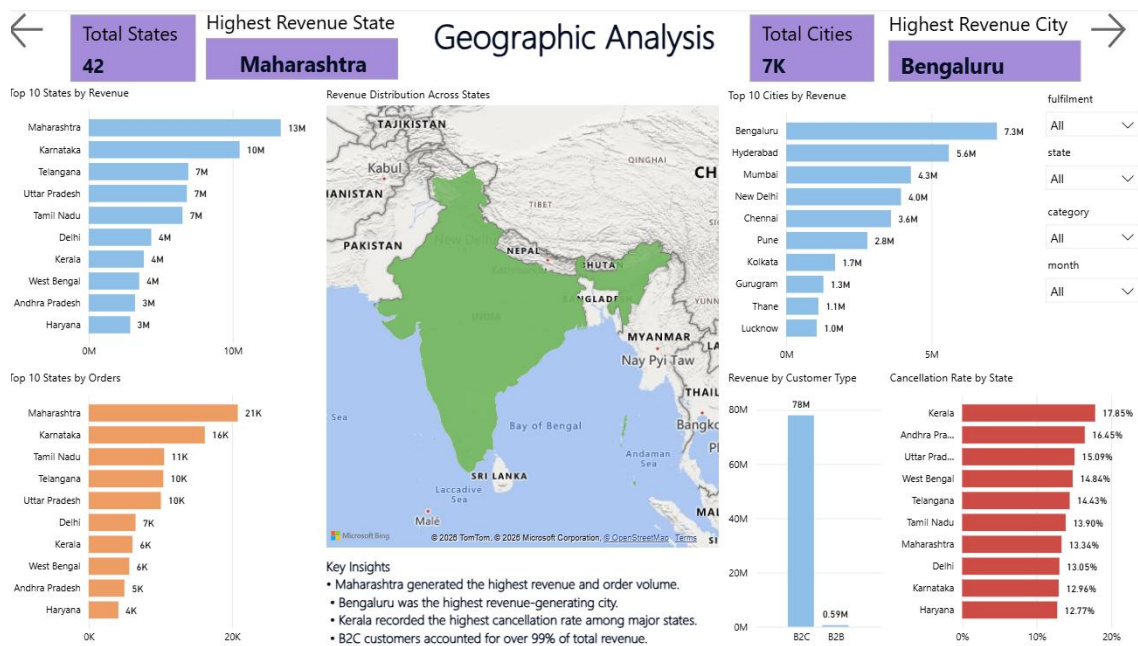


Figure 4. Geographic Analysis Dashboard developed in Power BI.

The Geographic Analysis dashboard examines sales performance across different states and cities in India. It provides geographical insights into customer demand, regional revenue contribution, and operational performance.

The dashboard includes:

- Top 10 states by revenue.
- Top 10 states by order volume.
- Top 10 cities by revenue.
- Revenue distribution displayed using a geographic map.
- State-wise cancellation rate comparison.
- Revenue distribution by customer type (B2B and B2C).
- KPI cards highlighting the highest revenue-generating state, highest revenue-generating city, total states, and total cities.

This dashboard helps identify high-performing markets, regions requiring operational improvements, and opportunities for business expansion.

9.3 Operational Analysis

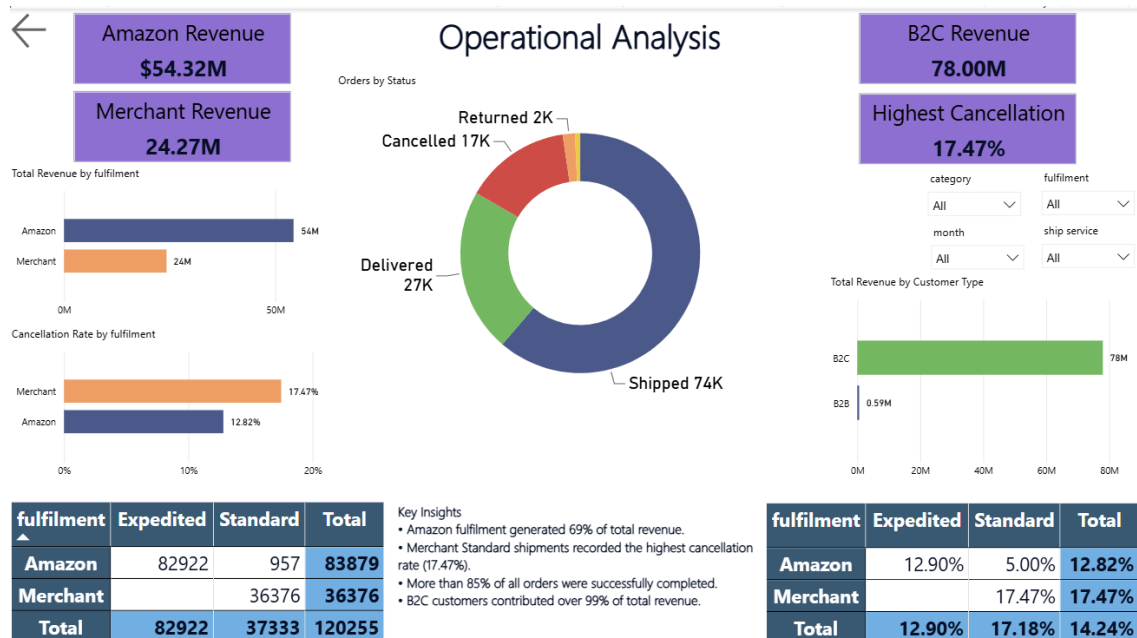


Figure 5. Operational Analysis Dashboard developed in Power BI.

The Operational Analysis dashboard focuses on order fulfilment performance and operational efficiency. It enables evaluation of fulfilment methods, shipping services, customer types, and order status distribution.

The dashboard includes:

- Revenue comparison between Amazon and Merchant fulfilment.
- Revenue distribution by customer type (B2B vs B2C).
- Order status distribution.
- Cancellation rate comparison by fulfilment method.
- Matrix visual summarizing fulfilment methods with shipping service levels.
- Matrix displaying cancellation rates across fulfilment and shipping combinations.
- KPI cards highlighting Amazon Revenue, Merchant Revenue, B2C Revenue, and Highest Cancellation Rate.
- Interactive slicers for Category, Month, Fulfilment, and Shipping Service.

The dashboard supports operational decision-making by identifying fulfilment methods with higher cancellation rates and comparing business performance across customer segments.

10. Key Learnings

This project provided practical experience in developing a complete end-to-end data analytics solution using industry-standard tools and techniques. The project strengthened both technical and business analytical skills while demonstrating the complete analytics lifecycle.

Technical Skills Acquired

- Data cleaning and preprocessing using Python (Pandas and NumPy).
- Data quality assessment, missing value handling, and data transformation.
- SQL query writing for KPI generation and business analysis.
- Database design and structured data validation.
- Creation of DAX measures and calculated columns in Power BI.
- Interactive dashboard development using charts, KPI cards, maps, slicers, and matrix visuals.
- Data visualization and dashboard design best practices.

Business & Analytical Skills

- Translating business questions into analytical queries.
- Identifying key performance indicators (KPIs).
- Product performance analysis.
- Geographic sales analysis.
- Fulfilment and operational performance evaluation.
- Time-series trend analysis.
- Converting analytical findings into actionable business recommendations.

This project enhanced problem-solving abilities and provided hands-on experience in building data-driven business intelligence solutions from raw transactional data.

11. Conclusion

This project demonstrates the complete end-to-end data analytics workflow, beginning with raw transactional data and ending with interactive business intelligence dashboards.

The dataset was first profiled, cleaned, and transformed using Python before being imported into SQL for business analysis. Multiple SQL queries were developed to evaluate executive KPIs, product performance, geographic trends, fulfilment operations, and time-series sales patterns. The analytical results were then visualized through interactive Power BI dashboards to support business decision-making.

The analysis identified key revenue-generating products, regions, customer segments, and fulfilment methods while highlighting operational challenges such as cancellation rates and regional

performance differences. Business recommendations were provided throughout the project to improve operational efficiency and support strategic decision-making.

Overall, this project demonstrates the ability to perform an end-to-end data analytics workflow, from raw data preprocessing and SQL-based business analysis to interactive dashboard development. It showcases practical proficiency in Python, SQL, Excel, DAX, and Power BI while emphasizing data-driven decision-making and business problem solving.

This project serves as a practical demonstration of end-to-end Data Analytics skills, covering data cleaning, SQL analysis, business intelligence, and dashboard development using Python, SQL, Excel, and Power BI.
